Scoring Combine-agent answers: RAGAS vs rubric vs key points

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RAGAS factual correctness disagrees with the rubric

`gmn-control-region-1`, mean of 3 runs per configuration

agent	FC recall	rubric (1–5)
GPT-5.6 luna, 3 configs	0.43–0.55	5.0
GPT-5.6 terra, 3 configs	0.41–0.66	5.0
Qwen 3.8, changed config	0.55	3.7

- 37 answers got rubric 5/5; their FC ranges 0.22 to 0.92
- reading them: the GPT answers are correct

What `FactualCorrectness(mode="recall")` computes
(ragas 0.2.15, `_factual_correctness.py`)

$$\text{recall} = \frac{TP}{TP + FN}$$

- *TP*: claims split from the **answer** that the gold supports
- *FN*: claims split from the **gold** missing in the answer
- two different LLM splits, redone on every call
- every gold claim weighs the same, side remarks included
  (e.g. the docs' "∼50%" comment in gmn)

Not "fraction of gold covered": it moves with answer length and claim split

Replacement: per-question key points ("nuggets")

Written once per question, frozen, versioned (`nuggets.yaml`):

`gmn-control-region-1`	
vital	use the `gmN` uncertainty type
vital	CR count N written after the keyword
vital	extrapolation factor α in the process column
vital	rate $= N\alpha$
okay	concrete datacard line; $1/\sqrt{N+1} \approx 45\%$
must_not	recommends `lnN`/`lnU`; example card inconsistent

Judge (Qwen 3.8 on aigw) per item:

- supported / partly / not supported
- must quote the answer; harness checks the quote is really there
- gold prose not shown; items shuffled; 3 judge runs, majority

Scores per answer:

$$V_{\text{strict}} = \frac{\#\,\text{vital fully supported}}{\#\,\text{vital}}$$

pass = all vital supported **and** no must_not present

Method of the TREC 2024/25 RAG track (AutoNuggetizer), plus evidence quotes and must_not items

Results: 2 questions × 8 agent configurations × 3 runs

question	agent / config	nugget pass	V_{strict}	FC recall	rubric
gmn	Qwen vanilla	0/3	0.00	0.00	1.0
	Qwen changed	2/3	0.67	0.55	3.7
	luna (3 configs)	3/3, 3/3, 3/3	1.00	0.43–0.55	5.0
	terra (3 configs)	2/3, 3/3, 3/3	1.00	0.41–0.66	5.0
shape-counting	Qwen vanilla	0/3	0.00	0.00	1.0
	Qwen changed	1/3	0.33	0.44	2.7
	luna van./v1/chg.	2/3, 1/3, 3/3	0.67, 0.50, 1.00	0.48, 0.60, 0.84	3.7, 3.7, 5.0
	terra van./v1/chg.	3/3, 3/3, 2/3	1.00, 1.00, 0.83	0.85, 0.80, 0.84	5.0

vanilla: shipped config (bash `ask`); v1: bash `deny` + read-only allow-list; changed: v1 + web fetch/search denied. Killed or empty runs count as 0.

- per answer: nugget pass agrees with "rubric ≥ 4" on 45/48 answers, with "FC ≥ 0.5" on 38/48
- correct GPT answers on gmn: $V_{\text{strict}} = 1.00$ where FC gave 0.41–0.55

What the nuggets add: a named reason for every failure

Caught, rubric said 5/5

- terra vanilla, gmn run 2: example card `bkg_stat gmN - 0.12` (no *N* after gmN) → must_not "inconsistent example" present

Diagnosed

- killed luna runs (permission reject, path typo): miss both vital points, not "low recall"
- Qwen changed, shape run 1: "shape and counting cards cannot be combined" → must_not

Missed / wrong (judge limits)

- terra changed, gmn run 2: `rate 10.0` with `gmN 25 0.12` ($25 \times 0.12 = 3.0$): arithmetic not caught
- 2 correct shape-counting answers failed: judge padded its quote with text not in the answer, quote check rejected it

rubric-vs-nugget disagreements	3/48
real error found by nuggets	1
judge false negative	2

Caveats and next steps

Caveats

- only 2 questions; key points written after seeing answers
- not yet calibrated against human labels
- Qwen judge grades Qwen answers (self-preference)
- 2 vital points on shape-counting: one judge flip $= \pm 0.5$
- agreement with humans is good for ranking systems, weak per question (TREC: τ 0.73–0.90 vs 0.3–0.66)

Next

- key points for all questions, drafted automatically from the **gold only** (LLM ensemble), then reviewed
- must_not items mined from observed wrong answers
- one-off calibration vs the 13 Oct. hand grades (Scott's π, judge leniency)
- cross-family judge for headline numbers
- report pass@1, pass^3, errors clustered by question

Proposal: nugget pass and V_{strict} as headline, rubric as second opinion, drop FC recall

References (scoring)

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